FasterCap 3D Input Files

Quadrilateral panel definitions ('Q' statement)

```
Syntax: Q <condname> <x1> <y1> <z1>   <x2> <y2> <z2>   <x3> <y3> <z3>   <x4> <y4> <z4>
[<xref> <yref> <zref>]
```

The 'Q' element at the beginning of a line defines a quadrilateral panel. If the quadrilateral panel belongs to a conductor surface, the `<condname>` parameter is used to associate the panel to a parent conductor named `<condname>`. If the quadrilateral panel belongs to a dielectric surface, the `<condname>` parameter is ignored. The *(x1, y1, z1)*, *(x2, y2, z2)*, *(x3, y3, z3)*, *(x4, y4, z4)* 3D points specify the coordinates of the four corners of the panel; the coordinates must be entered in clockwise or counterclockwise order starting from any one corner.
In case the panel is part of a dielectric interface, you can use the optional coordinates *(xref, yref, zref)* to specify a reference point; see 3D Dielectric definitions ('D' statement) for more details about dielectric interfaces.

Example 1:

```
Q mycube  1.0 1.0 0.0  1.0 0.0 0.0  1.0 0.0 1.0  1.0 1.0 1.0
```

This input file fragments specifies a face of the surface of a conducting cube, whose name is `mycube`.

Example 2:

```
Q onecube  1.0 1.0 0.0  1.0 0.0 0.0  1.0 0.0 1.0  1.0 1.0 1.0
Q othercube  0.0 1.0 0.0  1.0 1.0 0.0  1.0 1.0 1.0  0.0 1.0 1.0
Q onecube  1.0 0.0 0.0  0.0 0.0 0.0  0.0 0.0 1.0  1.0 0.0 1.0
```

This input file fragments specifies two face of the surface of a conducting cube, whose name is `onecube`, and one face of the surface of a second conducting cube, whose name is `othercube`.

Example 3:

```
Q mycube  1.0 1.0 0.0  1.0 0.0 0.0  1.0 0.0 1.0  1.0 1.0 1.0  0.0 0.0 2.0
```

This input file fragments specifies a face of the surface of a dielectric cube, whose name is `mycube`, with a panel-specific dielectric reference point at (0,0,2).